

QUESTION 1: VLSM for 5 Departments

A company has been allocated the network **192.168.50.0/24**. They have the following departments and host requirements:

Department Required Hosts

Admin	60
HR	30
Sales	28
IT	12
Security	6

Task:

Using **VLSM**, create subnets for each department and determine:

1. Subnet Mask (CIDR)
2. Network Address
3. Broadcast Address
4. 1st Valid Host
5. Last Valid Host

QUESTION 2 – CIDR Subnetting for 4 Branches

A company is assigned the block **10.10.0.0/22**.

This company wants to divide the address space into **4 equal-sized subnets** for its 4 branches:

- Head Office
- Lahore Branch
- Karachi Branch
- Islamabad Branch

Task:

Perform CIDR subnetting and provide for each subnet:

1. Subnet Number
2. Network Address
3. CIDR Notation
4. Broadcast Address
5. First Usable IP
6. Last Usable IP